

 Marwadi University Marwadi Chandarana Group		Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: Daily Expense app using getx.	
Experiment No:- 06	Date:- 18-10-2024	Enrollment No:- 92200133030	

Objective:- Daily Expense app using getx.

Code :-

➤ lib

• controller

○ expense_controller.dart

```
import 'package:expense_manager/model/expense_data.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';
import 'package:intl/intl.dart';

class ExpenseController extends GetxController {
  var selected = "All".obs;
  var curBottomNavIndex = 0.obs;

  var income = 0.0.obs;
  var spent = 0.0.obs;

  RxList<ExpenseData> expenses = <ExpenseData>[].obs;

  var category = "Food".obs;
  var paymentType = "Cash".obs;

  TextEditingController amountCntrl = TextEditingController();

  double amount = 0.0;

  List<List<dynamic>> iconList = [
    [Icons.home, "Home"],
    [Icons.swap_horiz, "Transfer"],
    [Icons.account_balance_wallet_outlined, "Wallet"],
    [Icons.account_circle_outlined, "Profile"],
  ];

  void saveData() {
    final newExpense = ExpenseData(
      amount: double.tryParse(amountCntrl.text) ?? 0,
      category: category.value,
      paymentType: paymentType.value,
      timestamp: DateFormat('MMM dd, yyyy').format(DateTime.now()),
    );

    expenses.add(newExpense);
    if (category.value == "Salary") {
      income.value += newExpense.amount;
    } else {
      spent.value += newExpense.amount;
    }
  }
}
```

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```

    }
  }
}

```

- **model**

- **expense_data.dart**

```

class ExpenseData {
  double amount;
  String category;
  String paymentType;
  String timestamp;

  ExpenseData({
    required this.amount,
    required this.category,
    required this.paymentType,
    required this.timestamp,
  });

  Map<String, dynamic> toMap() {
    return {
      'amount': amount,
      'category': category,
      'paymentType': paymentType,
      'timestamp': timestamp,
    };
  }
}

factory ExpenseData.fromMap(Map<String, dynamic> map) {
  return ExpenseData(
    amount: map['amount'],
    category: map['category'],
    paymentType: map['paymentType'],
    timestamp: map['timestamp'],
  );
}

```

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- screen
 - home_page.dart

```
import 'package:animated_bottom_navigation_bar/animated_bottom_navigation_bar.dart';
import 'package:charts_flutter/flutter.dart' as charts;
import 'package:expense_manager/controller/expense_controller.dart';
import 'package:expense_manager/screen/add_expense.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});
  static ExpenseController controller = Get.put(ExpenseController());

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Container(
        color: const Color(0xFFb6d8ae),
        child: SafeArea(
          child: Container(
            decoration: const BoxDecoration(
              color: Colors.white,
              borderRadius: BorderRadius.only(
                topLeft: Radius.circular(30),
                topRight: Radius.circular(30),
              ),
            ),
            padding: const EdgeInsets.fromLTRB(20, 20, 20, 0),
            child: Column(
              children: [
                Row(
                  children: [
                    const Expanded(
                      child: Column(
                        crossAxisAlignment: CrossAxisAlignment.start,
                        children: [
                          Text(
                            "Hello,",
                            style: TextStyle(fontSize: 24),
                          ),
                          Text(
                            "Aryan Langhanoja",
                            style: TextStyle(
                              fontSize: 24,
                              fontWeight: FontWeight.bold,
                            ),
                          ),
                        ],
                    ),
                  ],
                ),
                Container(
                  decoration: BoxDecoration(
```

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```

border: Border.all(color: Colors.grey[300]!),
borderRadius: BorderRadius.circular(32),
),
child: IconButton(
  onPressed: () {},
  icon: const Icon(Icons.notifications_none),
),
),
],
),
Obx(
  () => SingleChildScrollView(
    scrollDirection: Axis.horizontal,
    child: Row(
      children: [
        outlinedButton("All", () {
          controller.selected.value = "All";
        }),
        outlinedButton("Daily", () {
          controller.selected.value = "Daily";
        }),
        outlinedButton("Weekly", () {
          controller.selected.value = "Weekly";
        }),
        outlinedButton("Monthly", () {
          controller.selected.value = "Monthly";
        }),
        outlinedButton("Yearly", () {
          controller.selected.value = "Yearly";
        }),
      ],
    ),
  ),
),
Container(
  margin: const EdgeInsets.only(top: 20, bottom: 15),
  decoration: BoxDecoration(
    borderRadius: BorderRadius.circular(15),
    color: Colors.grey[100],
  ),
  child: Row(
    mainAxisAlignment: MainAxisAlignment.spaceAround,
    children: [
      incomeSpentSection(),
      pieChartSection(),
    ],
  ),
),
Row(
  children: [
    const Expanded(
      child: Text(

```



Aim: Daily Expense app using getx.

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```

        "Recent transactions",
        style: TextStyle(color: Colors.grey, fontSize: 20),
      ),
    ),
    OutlinedButton(
      onPressed: () {},
      style: OutlinedButton.styleFrom(
        padding: const EdgeInsets.fromLTRB(15, 0, 5, 0),
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(32),
        ),
        side: BorderSide(color: Colors.grey[300]!, width: 1.5),
      ),
      child: const Row(
        children: [
          Text(
            "See All",
            style: TextStyle(color: Colors.grey),
          ),
          Icon(
            Icons.chevron_right,
            color: Colors.grey,
          )
        ],
      ),
    ),
  ],
),
Expanded(
  child: Obx(
    () => ListView.builder(
      itemCount: controller.expenses.length,
      itemBuilder: (context, index) {
        final expense = controller.expenses[index];
        return expenseCard(
          expense.amount,
          expense.category,
          expense.paymentType,
          expense.timestamp,
        );
      },
    ),
  ),
),
],
),
),
),
floatingActionButton: FloatingActionButton(
  onPressed: () {
    Get.to(() => AddExpense());
  },
),

```

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```

    },
    backgroundColor: const Color(0xFFb7d9ae),
    child: Icon(
      Icons.add,
      color: Colors.grey[800],
    ),
  ),
  floatingActionButtonLocation: FloatingActionButtonLocation.centerDocked,
  bottomNavigationBar: Obx(
    () => AnimatedBottomNavigationBar.builder(
      backgroundColor: Colors.white,
      itemCount: controller.iconList.length,
      activeIndex: controller.curBottomNavIndex.value,
      onTap: (index) => controller.curBottomNavIndex.value = index,
      splashSpeedInMilliseconds: 300,
      notchSmoothness: NotchSmoothness.defaultEdge,
      gapLocation: GapLocation.center,
      tabBuilder: (int index, bool isActive) {
        return Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Icon(
              controller.iconList[index][0],
              size: 24,
              color: isActive ? const Color(0xFFb7d9ae) : Colors.grey[800],
            ),
            const SizedBox(height: 4),
            Padding(
              padding: const EdgeInsets.symmetric(horizontal: 8),
              child: Text(
                controller.iconList[index][1],
                style: TextStyle(color: Colors.grey[800]),
              ),
            ),
          ],
        ),
      ),
    ),
  ),
);
},
),
);
};
}

Widget outlinedButton(String label, VoidCallback onPressed) {
  return Padding(
    padding: const EdgeInsets.fromLTRB(0, 0, 10, 0),
    child: OutlinedButton(
      onPressed: onPressed,
      style: OutlinedButton.styleFrom(
        padding: const EdgeInsets.symmetric(horizontal: 20, vertical: 12),
        shape: RoundedRectangleBorder(
          borderRadius: BorderRadius.circular(15),
        ),
      ),
    ),
  ),
);
}

```

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```

        side: BorderSide(
          color: controller.selected.value == label
            ? const Color(0xFFb6d8ae)
            : Colors.grey[300]!,
          width: 1.5,
        ),
      ),
    child: Text(
      label,
      style: TextStyle(
        color:
          controller.selected.value != label ? Colors.grey : Colors.black,
      ),
    ),
  ),
);
}

Widget incomeSpentSection() {
  return Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      labelAndValue("Income", controller.income, const Color(0xFFb6d8ae)),
      const SizedBox(height: 20),
      labelAndValue("Spent", controller.spent, const Color(0xFFfeb9ab)),
    ],
  );
}

Widget labelAndValue(String label, RxDouble value, Color color) {
  return Column(
    children: [
      Row(
        children: [
          Container(
            width: 5,
            height: 15,
            decoration: BoxDecoration(
              borderRadius: BorderRadius.circular(3),
              color: color,
            ),
          ),
          Text(" $label"),
        ],
      ),
      Obx(
        () => Text(
          "₹${value.value % 1 == 0.0 ? value.value.toInt() :
value.value.toStringAsFixed(2)}",
          style: const TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
        ),
      ),
    ],
  );
}

```

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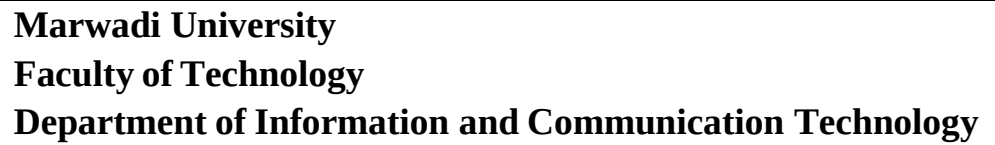
```

    ],
  );
}

Widget pieChartSection() {
  return Obx(
    () => SizedBox(
      width: 165,
      height: 165,
      child: charts.PieChart(
        [
          charts.Series<Expense, int>(
            id: 'Expense',
            domainFn: (Expense obj, _) => obj.label,
            measureFn: (Expense obj, _) => obj.value,
            data: [
              Expense(0, controller.income.value.toInt()),
              Expense(1, controller.spent.value.toInt()),
            ],
            colorFn: (datum, index) => [
              charts.ColorUtil.fromDartColor(const Color(0xFFb6d8ae)),
              charts.ColorUtil.fromDartColor(const Color(0xFFfeb9ab)),
            ][index!],
          ),
        ],
        animate: false,
        defaultRenderer: charts.ArcRendererConfig<int>(
          arcWidth: 30,
          startAngle: 31.4,
        ),
      ),
    ),
  );
}

Widget expenseCard(
  double amount, String category, String paymentType, String timestamp) {
  return Padding(
    padding: const EdgeInsets.fromLTRB(0, 0, 0, 15),
    child: Row(
      children: [
        Container(
          width: 48,
          height: 48,
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(15),
            color: category == "Food"
              ? const Color(0xFFfaedd4)
              : category == "Salary"
                ? const Color(0xFFe9f3e6)
                : const Color(0xFFf1e6ff),
          ),
        ),

```



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```

        child: Icon(
          category == "Food"
            ? Icons.emoji_food_beverage_outlined
            : category == "Salary"
              ? Icons.currency_rupee
              : Icons.local_movies_outlined,
          color: Colors.grey[800],
        ),
      ),
    const SizedBox(width: 15),
    Expanded(
      child: Column(
        children: [
          Row(
            children: [
              Expanded(
                child: Text(
                  category,
                  style: const TextStyle(fontWeight: FontWeight.bold),
                ),
              ),
              Text(
                "₹${amount % 1 == 0 ? amount.toInt() : amount}",
                style: const TextStyle(fontWeight: FontWeight.bold),
              ),
            ],
          ),
          Row(
            children: [
              Expanded(
                child: Text(
                  paymentType,
                  style: const TextStyle(color: Colors.grey),
                ),
              ),
              Text(
                timestamp,
                style: const TextStyle(color: Colors.grey),
              ),
            ],
          ),
        ],
      ),
    ),
  ],
),
),
),
),
),
),
),
);
}
}
}

```

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```
class Expense {
  final int label;
  final int value;

  Expense(this.label, this.value);
}
```

○ **add_expense.dart**

```
import 'package:expense_manager/controller/expense_controller.dart';
import 'package:flutter/material.dart';
import 'package:flutter/services.dart';
import 'package:get/get.dart';

class AddExpense extends StatelessWidget {
  AddExpense({super.key});
  final ExpenseController controller = Get.find<ExpenseController>();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: const Color(0xFFb6d8ae),
      body: SafeArea(
        child: Container(
          decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(30),
          ),
          padding: const EdgeInsets.fromLTRB(20, 30, 20, 30),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.start,
            children: [
              Row(
                children: [
                  OutlinedButton(
                    onPressed: () {},
                    style: OutlinedButton.styleFrom(
                      padding: const EdgeInsets.fromLTRB(10, 14, 10, 14),
                      shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(20),
                      ),
                      side: BorderSide(color: Colors.grey[300]!, width: 1.5),
                    ),
                    child: const Icon(
                      Icons.chevron_left,
                      color: Colors.grey,
                      size: 32,
                    ),
                  ),
                  SizedBox(width: MediaQuery.of(context).size.width * 0.09),
                  Text(
```

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```

        "Add transaction",
        style: TextStyle(fontSize: 24, color: Colors.grey[800]),
      ),
    ],
  ),
  Expanded(
    child: ListView(
      children: [
        Container(
          decoration: BoxDecoration(
            borderRadius: BorderRadius.circular(15),
            color: Colors.grey[100],
          ),
          padding: const EdgeInsets.symmetric(
            horizontal: 20, vertical: 10),
          margin: const EdgeInsets.fromLTRB(0, 20, 0, 5),
          child: TextField(
            controller: controller.amountCntrl,
            inputFormatters: [
              FilteringTextInputFormatter.allow(
                RegExp(r'^\d+\.\d{0,2}'),
                replacementString: ""),
            ],
            onTapOutside: (event) =>
              FocusScope.of(context).unfocus(),
            keyboardType: TextInputType.number,
            decoration: InputDecoration(
              border: InputBorder.none,
              label: Text(
                "Amount",
                style: TextStyle(color: Colors.grey[700]),
              ),
              alignLabelWithHint: false,
              prefixText: "₹",
              prefixStyle:
                TextStyle(color: Colors.grey[800], fontSize: 16),
            ),
          ),
        ),
      ],
    ),
    Container(
      width: double.infinity,
      decoration: BoxDecoration(
        borderRadius: BorderRadius.circular(15),
        color: Colors.grey[100],
      ),
      padding: const EdgeInsets.symmetric(
        horizontal: 20, vertical: 20),
      margin: const EdgeInsets.fromLTRB(0, 5, 0, 5),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text(

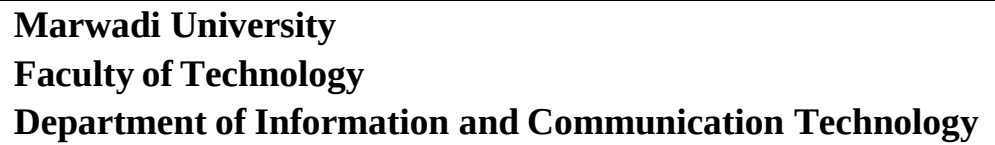
```

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```

"Category",
style: TextStyle(color: Colors.grey[700]),
),
const SizedBox(height: 10),
Obx(
() => DropdownButton(
value: controller.category.value,
items: [
dropdownCard(Icons.emoji_food_beverage_outlined,
"Food", const Color(0xFFfaedd4)),
dropdownCard(Icons.currency_rupee, "Salary",
const Color(0xFFe9f3e6)),
dropdownCard(Icons.local_movies_outlined,
"Entertainment", const Color(0xFFFF1e6ff)),
],
icon: const Icon(Icons.keyboard_arrow_down),
isExpanded: true,
underline: const SizedBox(),
onChanged: (value) {
controller.category.value = value!;
// setState(() => controller.category = value!);
},
),
),
),
),
const SizedBox(height: 20),
Text(
"Payment Type",
style: TextStyle(
fontSize: 24,
fontWeight: FontWeight.bold,
color: Colors.grey[800],
),
),
const SizedBox(height: 20),
Obx(
() => Column(
crossAxisAlignment: CrossAxisAlignment.start,
children: [
radioButtonCard("Cash", 130),
radioButtonCard("Credit/Debit Card", 220),
radioButtonCard("Check", 140),
],
),
),
),
),
Row(
children: [

```



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```
Expanded(
  child: OutlinedButton(
    onPressed: () {},
    style: OutlinedButton.styleFrom(
      padding: const EdgeInsets.symmetric(vertical: 20),
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(20),
      ),
    ),
    child: Text(
      "Draft",
      style: TextStyle(
        fontSize: 18,
        fontWeight: FontWeight.w600,
        color: Colors.grey[800]),
    ),
  ),
),
const SizedBox(width: 20),
Expanded(
  child: OutlinedButton(
    onPressed: () {
      controller.saveData();
      Get.back();
    },
    style: OutlinedButton.styleFrom(
      backgroundColor: const Color(0xFFb6d8ae),
      padding: const EdgeInsets.symmetric(vertical: 20),
      shape: RoundedRectangleBorder(
        borderRadius: BorderRadius.circular(20),
      ),
    ),
    child: Text(
      "Add",
      style: TextStyle(
        fontSize: 18,
        fontWeight: FontWeight.w600,
        color: Colors.grey[800]),
    ),
  ),
),
),
],
),
],
),
),
),
);
}
```

```
DropDownMenuItem dropDownCard(IconData icon, String label, Color backColor) {
    return DropDownMenuItem(
```

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```

value: label,
child: Row(
  children: [
    Container(
      width: 42,
      height: 42,
      decoration: BoxDecoration(
        borderRadius: BorderRadius.circular(15),
        color: backColor,
      ),
      child: Icon(
        icon,
        color: Colors.grey[800],
        size: 20,
      ),
    ),
    const SizedBox(width: 15),
    Text(
      label,
      style: TextStyle(
        fontSize: 18,
        color: Colors.grey[800],
        fontWeight: FontWeight.bold),
    ),
  ],
),
);
}

Widget radioButtonCard(String label, double width) {
  return Container(
    decoration: BoxDecoration(
      border: Border.all(
        color: controller.paymentType.value == label
          ? const Color(0xFFb6d8ae)
          : Colors.grey,
        width: 1.5,
      ),
      borderRadius: BorderRadius.circular(20),
    ),
    width: width,
    margin: const EdgeInsets.fromLTRB(0, 4, 0, 4),
    child: RadioListTile<String>(
      title: Text(
        label,
        style: TextStyle(
          color: controller.paymentType.value == label
            ? Colors.grey[800]
            : Colors.grey[600],
          fontSize: 16,
          fontWeight: FontWeight.bold),
      ),

```

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```

        value: label,
        groupValue: controller.paymentType.value,
        activeColor: const Color(0xFFb6d8ae),
        contentPadding: const EdgeInsets.symmetric(horizontal: 10),
        onChanged: (value) {
          controller.paymentType.value = value!;
        },
      ),
    );
  }
}

```

- **main.dart**

```

import 'package:expense_manager/screen/home_page.dart';
import 'package:flutter/material.dart';
import 'package:get/get.dart';

void main() {
  runApp(const MyApp());
}

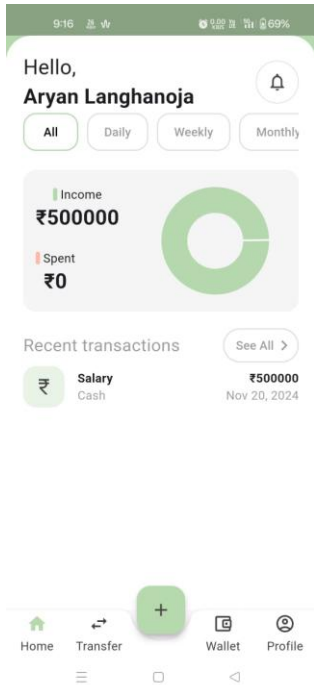
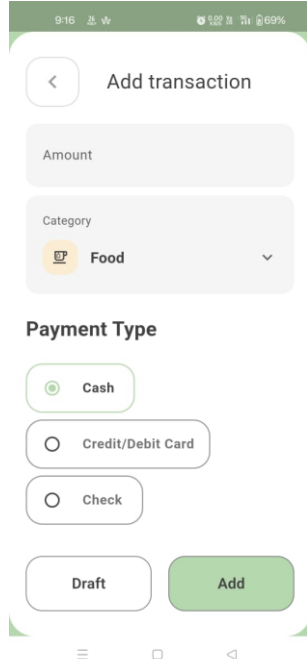
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return GetMaterialApp(
      title: 'Flutter Demo',
      theme: ThemeData(
        // colorScheme: ColorScheme.fromSeed(seedColor: Colors.deepPurple),
        // useMaterial3: true,
      ),
      debugShowCheckedModeBanner: false,
      home: const HomePage(),
    );
  }
}

```

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Output:-

9:16 69%

< Add transaction

Amount

Category **Food**

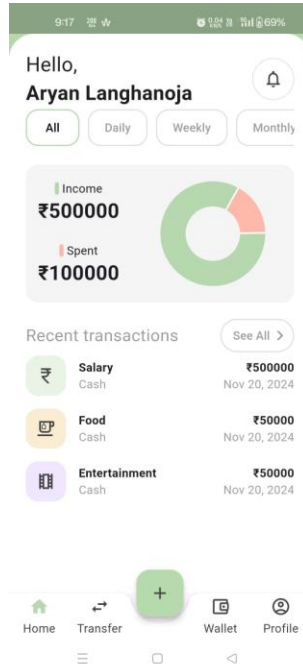
Payment Type

☒ Cash

☐ Credit/Debit Card

☐ Check

Draft Add



 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Cross-Platform Mobile Application Development (01CT1517)		Aim: Daily Expense app using getx.	
Experiment No:- 06	Date:- 18-10-2024	Enrollment No:- 92200133030	